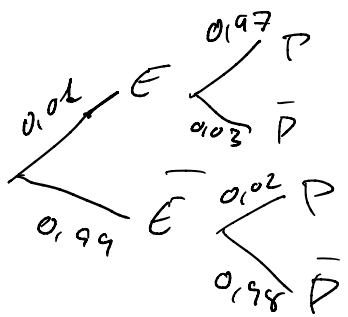
Estadística

Refuerzo Juan

Made with Goodnotes

Ej 3

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$



a) $P(P \cap E) = 0,01 \cdot 0,97$
$= 0,0097$

b) $P(E|P) = \frac{P(E \cap P)}{P(P)}$

$P(P) = 0,01 \cdot 0,97$
$+ 0,99 \cdot 0,02 = 0,0295$

$P(E|P) = \frac{0,0097}{0,0295} = 0,329$

Ej 4

A = Asistencia a clase:

$P(\bar{A}) = ?/3$ $\quad P(A|\bar{S}) = P(A|S) = 0,5$

$P(\bar{S}|\bar{A}) = 0,75$ $\quad P(A|\bar{B}) = ??$

$$P(A|\bar{S}) = \frac{P(A \cap \bar{S})}{P(\bar{S})} = \frac{P(A \cap S)}{P(S)}$$

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Ej 1 -> Parcial MAIS 2023

$P(A) = 0,6$ $\quad P(T_1) = 0,7$ $\quad$ C = corte comunicação

$P(B) = 0,4$ $\quad P(T_1 \cap T_2) = 0,3$ $\quad P(T_1 \cup T_2) = 1$

$P(C|A) = 0,1$ $\quad P(C|B) = 0,15$ $\quad P(C|T_1) = 0,05$

a) $P(C) = ??$

$$P(C|A) = \frac{P(C \cap A)}{P(A)} \Rightarrow P(C \cap A) = 0,06$$

$$P(C|B) = \frac{P(C \cap B)}{P(B)} = \frac{P(C) - P(C \cap A)}{P(B)}$$

$0,15 \cdot 0,4 = P(C) - 0,06 \quad = 0,12$

b) $P(T_2 \cap \overline{T_1}) = P(T_2) - P(T_2 \cap T_1)$

$P(T_1) + P(T_2) - P(T_1 \cap T_2) = P(T_1 \cup T_2)$

$0,7 + P(T_2) - 0,3 = 1 \rightarrow P(T_2) = 0,6$

$P(T_2 \cap \overline{T_1}) = 0,6 - 0,3 = 0,3$

c) $P(T_1|C) = \frac{P(T_1 \cap C)}{P(C)}$ $\quad P(C|T_1) = \frac{P(T_1 \cap C)}{P(T_1)}$

$P(T_1|C) = \frac{0,035}{0,12} = \frac{7}{24} = 0,292$ $\quad P(T_2 \cap C) = 0,05 \cdot 0,7 = 0,035$

d) $P(C|\overline{T_1}) = \frac{P(C \cap \overline{T_1})}{P(\overline{T_1})} = \frac{P(C) - P(C \cap T_1)}{P(\overline{T_1})} = 0,283$

$P(\overline{T_1}) = 1 - P(T_1) = 0,3$

Made with Goodnotes

Ej 7

$X \sim N(70,3)$

a) $P(67 < x < 75) = P(x < 75) - P(x < 67)$

$= P\left(z < \frac{75-70}{3}\right) - P\left(z < \frac{67-70}{3}\right)$

$= P(z < 1,67) - P(z < -1) =$

$0,95 - 0,1587 = 0,794$

$P(R) = 1 - 0,794 = 0,206$

b) $P(x < 73,84) = P\left(z < \frac{73,84-70}{3}\right)$

$= P(z < 1,28) = 0,8997$

$Y \sim B(10; 0,8997)$

$P(Y \leq 8) = 1 - P(Y=10) - P(Y=9)$

$= 1 - \binom{10}{10} 0,8997^{10} (0,1003)^0$

$- \binom{10}{9} \cdot 0,8997^9 \cdot 0,1003^1$

$= 0,265$

c) $x' \sim N(np, \sqrt{np(1-p)})$

$x' \sim N(89,97; 3,003)$

$P(x \leq 80) = P(x < 80,5) = P\left(z < \frac{80,5-89,97}{3,003}\right)$

$= P(z < -3,15) = 8,06 \cdot 10^{-4}$

Made with Goodnotes

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